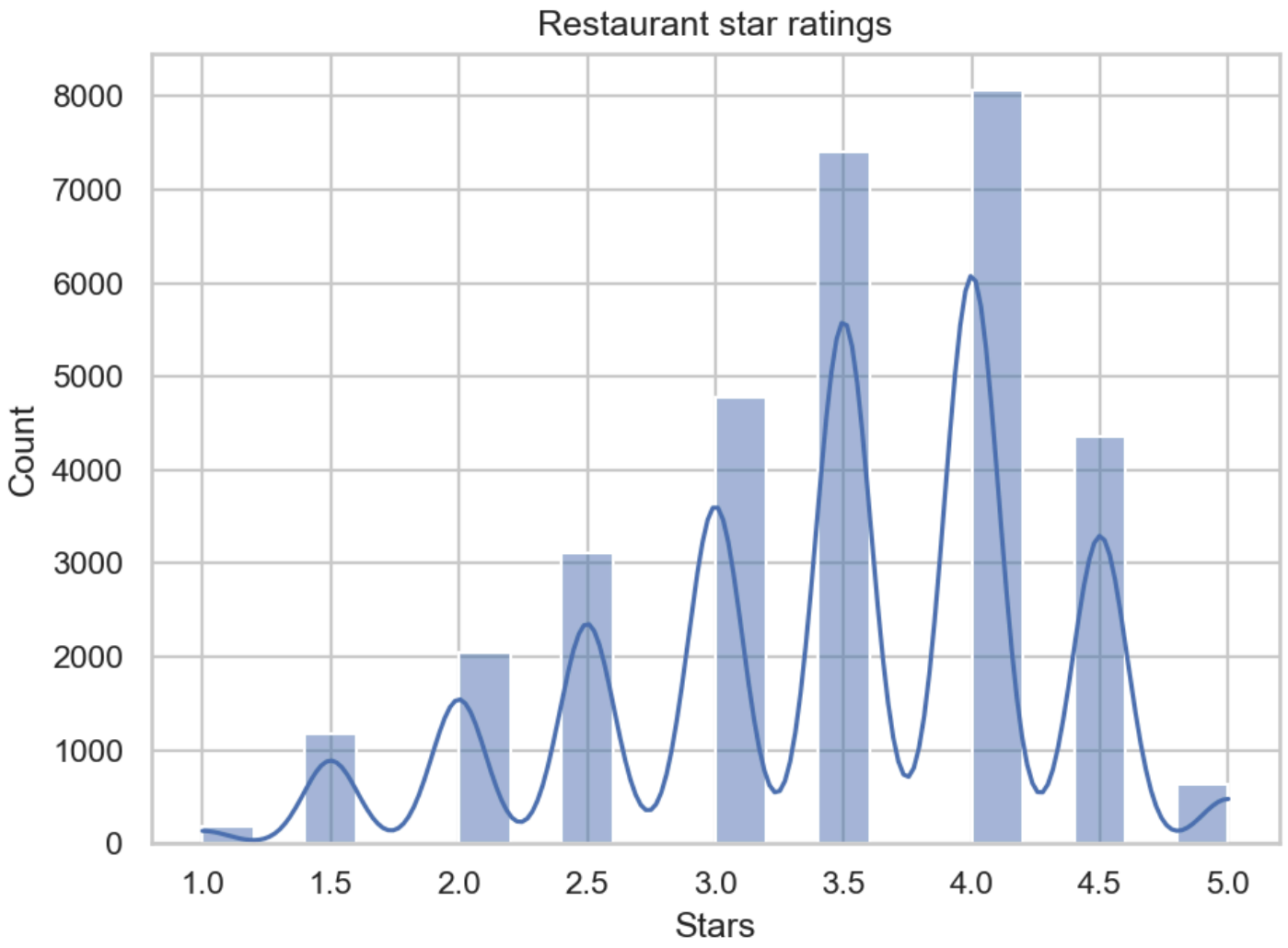
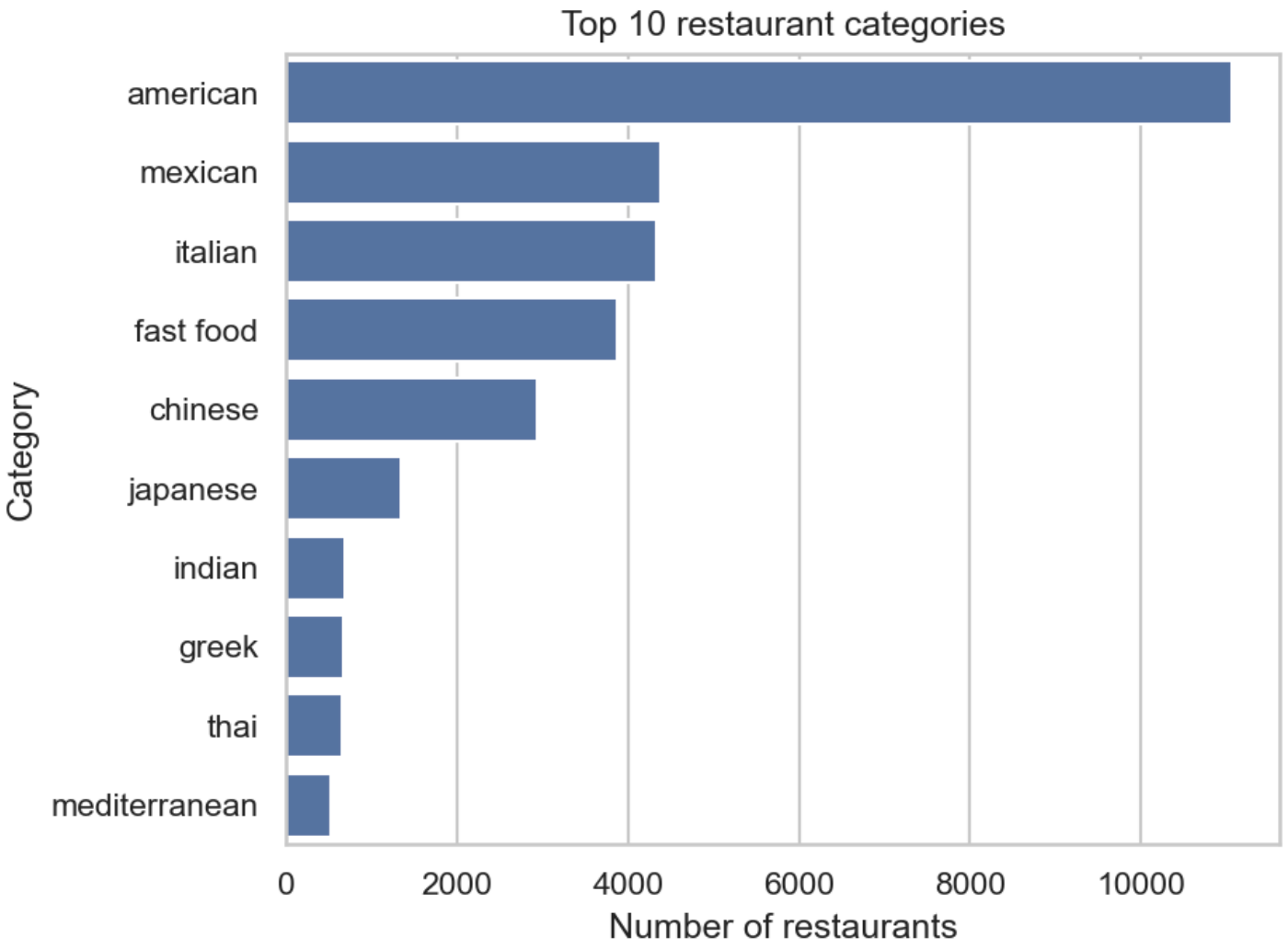
- Restaurant filtering: Categories are first filled with `fillna("")` to convert to lowercase, filtering for records containing "restaurant"; restaurants without a recognized cuisine are discarded.
- Dimension tables: `category_list` is dropped with `dropna()` and then `dim_category` is created; missing/empty elites in `dim_user` are treated as 0 years; `dim_date` derives its date key from the review date.
- Text cleaning: Missing entries are skipped during name/address normalization, keeping NA values.
- Fact table alignment: After cleaning `business_id` in `fact_review`, only records matching the restaurant dimension are kept, and mismatched IDs are printed.
- City key: After merging `city_key`, the number of NULL rows is counted, and no padding is performed.

Visualization of the data

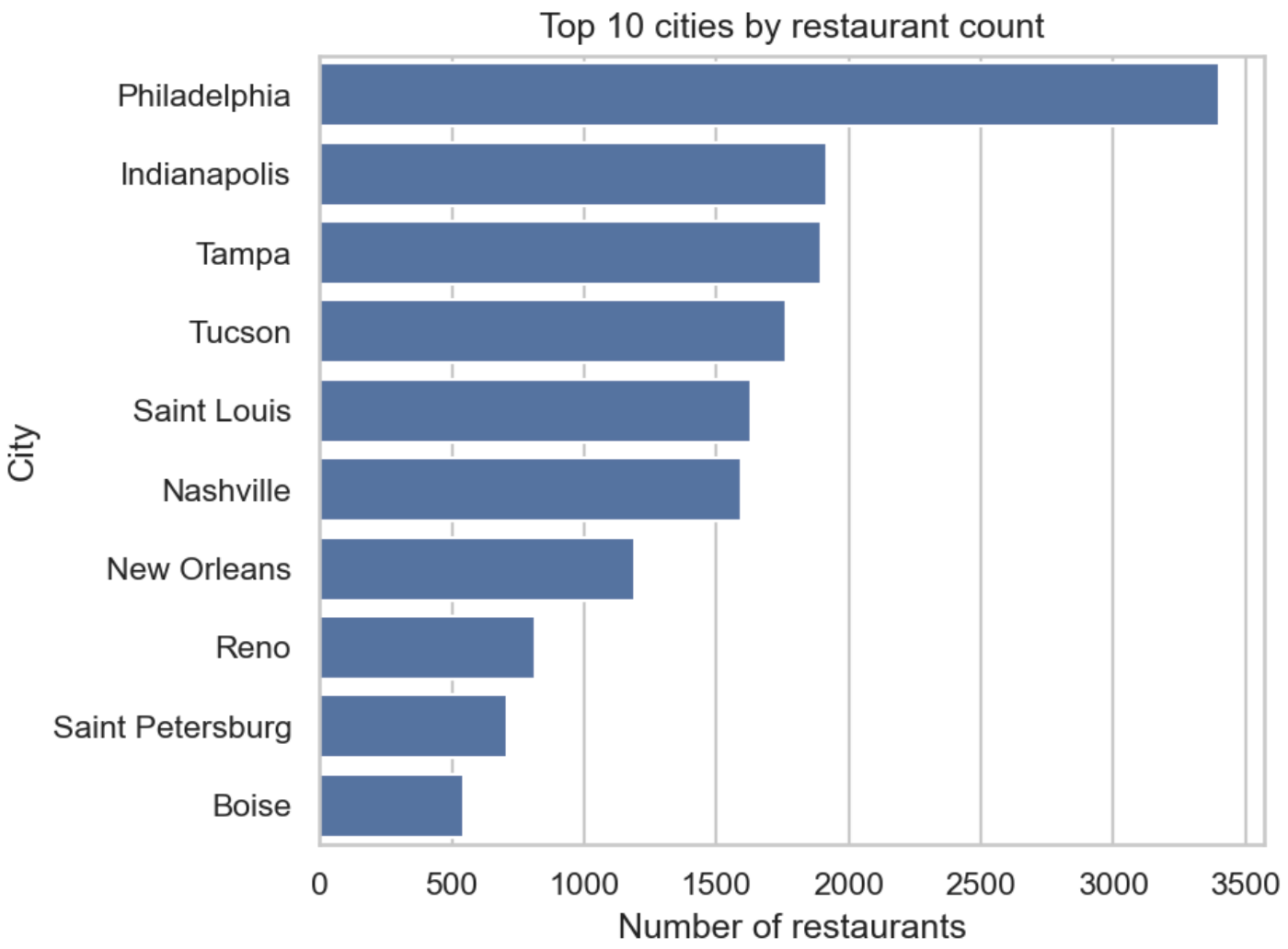


The main peaks are in the range of 3.0, 3.5, 4.0, and 4.5, indicating that most restaurants are concentrated in the mid-to-high score range, especially in the area of 3.5–4.0.

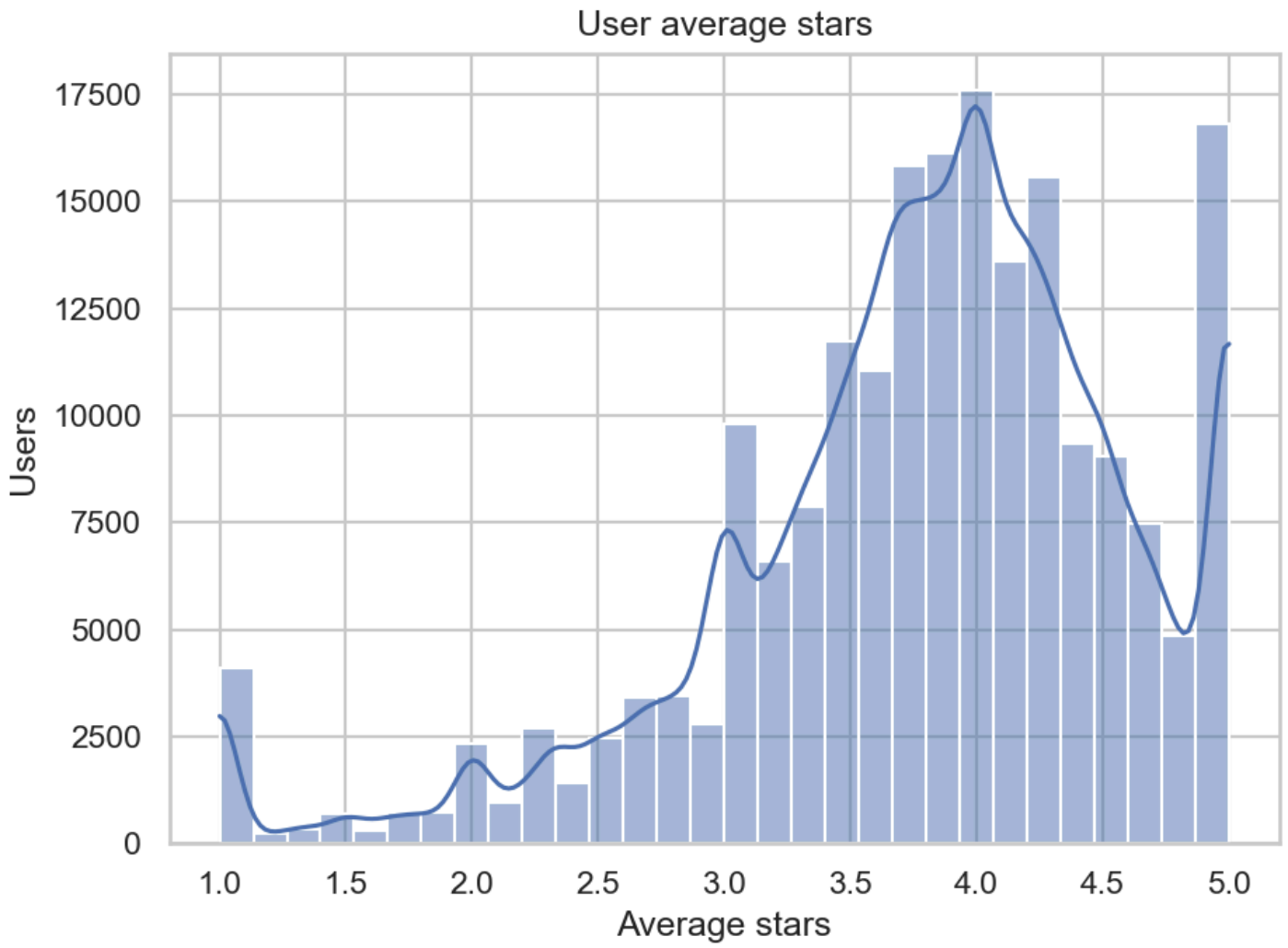


The horizontal axis represents the number of restaurants, and the vertical axis represents the cuisine category, listing the top 10 within the sample.

- The largest category is American (approximately 11,000 restaurants), significantly exceeding other categories.
- The market structure is highly concentrated in American restaurants, followed by Mexican/Italian/fast food, with a rapid decline below that, and specialty/niche cuisines accounting for a very small percentage.



- Horizontal axis: Number of restaurants; Vertical axis: City name.
- Restaurant distribution is highly concentrated in a few large cities, especially Philadelphia; the cities on the list also represent a more competitive market.



User average star rating distribution.

- Horizontal axis: Average star rating (1–5) for all reviews by each user.
- Vertical axis: Number of users within the corresponding range.
- The main peak is around 3.7–4.2, indicating that most users generally give relatively high ratings and are quite lenient. The mean is approximately 3.77, and the 50th percentile is approximately 3.88.
- There is also a noticeable 5-star column/peak on the right, usually caused by users with very few reviews and all ratings of 5 (or users with optimistic ratings).
- The percentage on the left (≤2.5) is very small, indicating that there are fewer strict/low-rated users.